Notes from Hatcher - Algebraic Topology

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1 Basic definitions

We start by listing some very basic definitions necessary for the rest of the book.

1.1 Basic assumed topolgoical notions

Here are a few absolutely elementary topological definitions, that are written down for good measure.

The first is the product topology. It is easy to remember what it is, just by thinking about the categorical definition of a product (as an inverse limit).

Definition 1.1.1: Product topology

Let $\{X_i\}$ be a family of spaces, and let $X := \prod_i X_i$ be the cartesian product of these spaces. The **product topology** on X is the coarsest topology on X such that all the projection maps $p_i : X \to X_i$ are continuous. Equivalently, it is given by the basis $\{p_i^{-1}(U_{ij})\}$, where U_{ij} runs over the open sets $U_{ij} \subseteq X_i$. Again, in other words, the open sets of X is given by the (possibly infinite) union of sets of the form $\prod_i U_i$, where $U_i \neq X_i$ for only finitely many i.

Similarily, the definition of a quotient topology, is pretty much the only thing it can be.

Definition 1.1.2: Quotient topology

Let X be a set, and $\sim$ an equivalence relation on X. Define $Y := X/\sim$, and let $p : X \to Y$ be the canonical projection. The **quotient topology** on Y is defined such that $U \subseteq Y$ is open if and only if $p^{-1}(U)$ is open.

There are more categorical ways to define the quotient topololology as well: For any surjection $q : X \to Y$, we can define the quotient topology on Y as the **final topology** wrt. q: this is the finest topology on Y, that makes q continuous (in general, the final topology is essentially the topology that appears when taking direct limits). It has the universal property that

$$f : Y \to Z \text{ continuous} \Leftrightarrow f \circ g : X \to Z \text{ continuous}$$

The dual notion of the final topology is the initial topology, of which the product topology was an example (not surprising, since that would be an inverse limit).

We also recall that a space is said to be **connected** if it is not possible to write as a disjoint union of two or more non-empty open subsets. A stronger notion is that of a **path-connected** space, in which every point has a path connecting them. A space is **locally path-connected** if every point $x \in X$ admits arbitrarily small neighborhoods (i.e. for every open $x \in U$, there exists a $x \in V \subset U$) that are path-connected. Both path connected and the locally path connected assumption may be necessary, since they actually don't imply each other (but the spaces that are path-connected but not locally path connected are pretty wacky).

Retractions, Deformation retracts.

2 The Fundamental Group

2.1 Basic constructions

We recall the definition of the first invariant of a space we study, the fundamental group.

Definition 2.1.1: The Fundamental Group

Let (X, x_0) be a pointed space. The fundamental group $\pi_1(X, x_0)$ is defined as the homotopy classes $[f]$ of loops in x_0, under composition of paths.

Proposition 2.1.2

Let h be a path in X from x_0 to x_1. Then h induces an isomorphism

$$\beta_h : \pi_1(X, x_0) \to \pi_1(X, x_1).$$

Proof. Just do the only thing you can do: Compose loops with h to and from. Everything then follows again since we only care about paths up to homotopy. □

Because of this, we often omit the base point x_0 from notation when X is path-connected.

By the above, the following definition is independent of base point. The following definition can also be described to say that every loop in X is contractible (nullhomotopic).

Definition 2.1.3

A path-connected space X is said to be **simply connected** if $\pi_1(X, x_0)$ is trivial for one choice (and hence all choices) of base-point x_0.

The first goal of this chapter is then to prove $\pi_1(S_1) \cong \mathbb{Z}$. This is done by introducing the quite important and general notion of a covering space.

Definition 2.1.4: Covering Space

A **covering space** $(\tilde{X}, p)$ of X is a space $\tilde{X}$ and a map $p : \tilde{X} \to X$ such that every point $x \in X$ has an open neighborhood $U_x \subset X$ such that $p^{-1}(U_x)$ is the disjoint union $\sqcup V_x$ of spaces V_x that are all mapped homeomorphically onto U_x by p. Moreover, such an U_x is said to be **evenly covered**.

The first example of a covering space, which will be used to compute the fundamental group of S^1 is the map

$$\begin{aligned} p : \mathbb{R} &\to S^1 \\ r &\to (\cos 2\pi r, \sin 2\pi r). \end{aligned}$$

Covering spaces are discussed in more detail in Section 2.3. Their lifting properties and similar will be discussed there. Assuming these lifting properties, the book then goes on to prove the structure of $\pi_1(S^1)$, and then prove numerous cool consequences.

It is clear that the construction of π_1 is in fact functorial: continuous maps between pointed spaces $s : (X, x_0) \to (Y, y_0)$ clearly induces homomorphisms $\varphi_s : \pi_1(X, x_0) \to \pi_1(Y, y_0)$, and identities and associativity conditions are trivial.

The following will imply the easy part of Van Kampen's theorem, but we state it here, as the book.

Lemma 2.1.5

Let X be a space, and let $\{A_\alpha\}_\alpha$ be path-connected open sets that cover X, with each open containing the basepoint $x_0 \in A_\alpha$. If each intersections $A_\alpha \cap A_\beta$ are all path-connected, then each loop in X based at x_0 is homotopic to a product of loops, all contained in a single A_α.

Proof. The idea is very simple: a loop f can be partitioned into pieces all contained in a single A_α. Between each partition, we are in an intersection, and we can then go back and forth between the base-point x_0. □

A nice consequence is already the following

Proposition 2.1.6

$\pi_1(S^n) = 0$ for $n \geq 2$.

Proof. Let A_1, A_2 be the compliments of two antipodal points on S^n. Then $A_i \cong \mathbb{R}^n$, and $A_1 \cap A_2 \cong \mathbb{R} \times S^{n-1}$, which is path-connected for $n \geq 2$. Thus, for any basepoint $x_0 \in A_1 \cap A_2$, Lemma 2.1.5 implies that every loop f is homotopic to a product of loops in A_1 and A_2 separately, but $\pi_1(A_i) = 0$, since they are both homeomorphic to $\mathbb{R}^n$, and hence f must be nullhomotopic. □

2.2 Van Kampen

The Van Kampen theorem is probably the first major theorem in the book, and allows us to compute fundamental groups from sufficiently nice covers.

Theorem 2.2.1: Van Kampen

Let X and A_α be as in Lemma 2.1.5. If each intersection $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\Phi : *_\alpha \pi(A_\alpha, x_0) \to \pi(X, x_0)$ is surjective. If additionally all the triple intersections $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then the kernel of Φ is the normal subgroup generated by the elements of the form $\iota_{\alpha\beta}(\omega)\iota_{\beta\alpha}(\omega)^{-1}$, where $\iota_{\alpha\beta}$ denotes the homomorphism $\pi_1(A_\alpha \cap A_\beta, x_0) \to \pi_1(A_\alpha)$ induced by the inclusion.
Specializing to the case where there are only two open sets A_1, A_2, the theorem states that when $A_1 \cap A_2$ is path-connected, $\pi_1(X, x_0)$ is exactly the pushout (in the category of groups)

$$\begin{array}{ccc} \pi_1(A_1 \cap A_2, x_0) & \longrightarrow & \pi_1(A_2, x_0) \\ \downarrow & & \downarrow \\ \pi_1(A_1, x_0) & \longrightarrow & \pi_1(X, x_0) = \pi_1(A_1, x_0) *_{\pi_1(A_1 \cap A_2, x_0)} \pi_1(A_2, x_0) \end{array}$$

Proof. It is easy to see that Φ is a homomorphism (induced by the inclusions $A_\alpha \hookrightarrow X$). Then, the surjectivity part is just Lemma 2.1.5.

To describe the kernel, we can choose a "factorization" of a loop $[f] \in \pi_1(X, x_0)$, (i.e. write f as a composition of loops all contained in single A_α's), and then show that all such choices give the same element once we quotient out by the relations describing the kernel of Φ. The idea of the proof of this is to consider a homotopy $I \times I \to X$ between two factorizations, and subdivide this square into a bunch of rectangles, each contained in a single A_α. Then one sees that moving across the rectangles may only change the factorizations by one of the relations. □

There are many good applications of Van Kampen in the book. A particularly nice one is the computation of the fundamental group of a cell-complex.

Proposition 2.2.2

Let Y be a space obtained from X by attaching 2-cells e^2_α along $\varphi_\alpha : S^1 \to X$. The loops defined by φ_α (this involves choosing a path to the basepoint x_0 to any point in this loop) are all clearly null-homotopic in Y, and thus lie in the kernel of the map $\varphi_i : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ induced by the inclusion. Denote the normal subgroup generated by all these loops by N. Then:

- φ_i is surjective with kernel N.
- If Y is instead obtained from X by attacking n-cells with $n > 2$, then φ_i is an isomorphism.
- For a path connected cell-complex, the inclusion of the 2-skeleton $X^2 \hookrightarrow X$ induces an isomorphism $\pi_1(X^2, x_0) \to \pi_1(X, x_0)$.

Proof. The proof of a) is quite nice and visual, see the book. Proof of b) is the same as a), but with a different conclusion. And finally c) follows easily from b) (at least in the finite-dimensional case, but the infinite dimensional case need to take a bit more care, but is still true). □

One nice corollary of the proposition above is essentially a solution to the topological version of the inverse Galois problem (the relation between galois theory and the fundamental group becomes a bit more clear in the next section, and rigorous in a later note on Étale cohomology).

Corollary 2.2.3

Let G be a group. There exists a 2-dimensional cell complex X_G with $\pi_1(X_G) \cong G$.

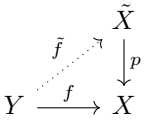
Proof. Write G from its possibly infinite presentation $\langle g_\alpha \mid r_\beta \rangle$ (i.e. g_α being the free generators, and r_β the words defining the relations). Then form X_G by starting with $\bigvee_\alpha S_\alpha$, and attaching 2-cells e^2_β by the loops defined by the relations r_β. □

2.3 Covering spaces

In this section, we describe covering spaces, whose definition was given eawrlier, in much more detail. The main goal of the section is to show the Galois-correspondence between covering spaces and the fundamental group.

Throughout this section, for a space X, $\tilde{X}$ will denote a covering space, and $p : \tilde{X} \to X$ will denote the corresponding covering map.

The raison d'etre of covering spaces is their lifting properties. Recall that given a map $f : Y \to X$, a lift of f is a map $\tilde{f}$ such that the following diagram commutes



The first lifting property is the **homotopy lifting property**.

Proposition 2.3.1: Homotopy lifting property

Given a homotopy $f_t : Y \to X$, and a map $\tilde{f}_0$ lifting f_0, there exists a unique homotopy $\tilde{f}_t : Y \to \tilde{X}$ of $\tilde{f}_0$ lifting f_t. In particular, taking Y to be a point, we get the **path lifting property**, that for every path $f : I \to X$, and choice of lift $\tilde{x_0}$ of $x_0 = f(0)$, there exists a unique path $\tilde{f} : I \to \tilde{X}$.

Proof. The proof is done in three steps: For a given point $y \in Y$, we can construct a neighborhood N and a lift of $F : N \times I \to X$. This is done by segmenting I into segments that are evenly covered. By compactness of I, this requires only finitely many segments.

Next, one shows that such a lift is unique for each point (this is direct from the definition of a covering space), and thus the lifts constructed in the first step glue together. □

There is immidiately a nice application of the homotopy lifting property in the following

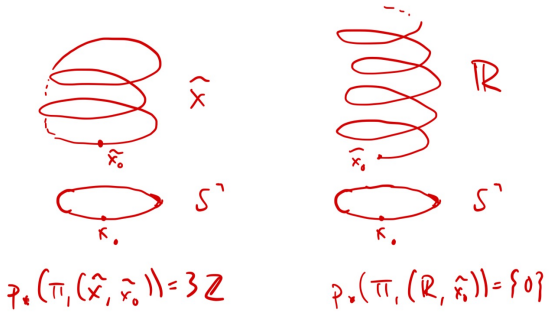
Proposition 2.3.2

The map $p_* : \pi_1(\tilde{X}, \tilde{x_0}) \to \pi_1(X, x_0)$ induced by a covering space is injective. The image $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ consists of homotopy classes of loops starting in x_0 whose lifts become loops in $\tilde{X}$ starting in $\tilde{x_0}$.
Assuming $X, \tilde{X}$ are both path-connected, the number of sheets of the covering space $p : (\tilde{X}, \tilde{x_0}) \to (X, x_0)$ is equal to the index of $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ in $\pi_1(X, x_0)$.

Proof. By the homotopy lifting property, any loop that is null-homotopic in (X, x_0) would also give rise to a null-homotopy of the original loop. For the second statement, it is clear that such loops are in the image. However, any loop in $p_*(\pi_1(\tilde{X}, \tilde{x_0}))$ must be at least homotopic to such a loop, but by homotopy-lifting, it must also lift to a loop starting in $\tilde{x_0}$.

For the last statement, let $H = p_*(\pi_1(\tilde{X}, \tilde{x_0}))$, and notice that there is a well-defined function from the cosets of H to $p^{-1}(x_0)$, by the previous paragraph, sending a coset $H[g]$ to $g(1)$. This function is clearly injective, as if $g_1(1) = g_2(1)$, then we have $g_1\overline{g_2} \in H$, and it is surjective by the path-connectedness of $\tilde{X}$. □

The image of a covering space can be intuitively visualized by the following pretty ugly-looking drawing. It is also easy to see the statement about the sheets.



We now state the next two important lifting properties,

Proposition 2.3.3: The lifting criterion

Given a map $f : (Y, y_0) \to (X, x_0)$, where Y is path-connected and locally path-connected, a lift $\tilde{f} : (Y, y_0) \to (\tilde{X}, \tilde{x_0})$ exists if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\tilde{X}, \tilde{x_0})$.

Proof. The only if part is obvious, since $p\tilde{f} = f$. The idea for the other direction, is to define $\tilde{f}(y)$ by taking a path α from y_0 to y, and lifting $f\alpha$ and taking the end-point. The assumption is then necessary to get well-definedness. □

The final property is the following

Proposition 2.3.4: Unique lifting property

Assume Y is connected. If two lifts $\tilde{f}_1, \tilde{f}_2 : Y \to \tilde{X}$ of a map $f : Y \to X$ agree on a single point, then they agree on all of Y.

Proof. The idea is to show that both the set of points that they do and don't agree must be open. Thus The set of points where they agree is both open and closed, and connectedness then implies it must be either empty or all of Y. □

Next, we move on to the classification of covering spaces, which will bear a strong resemblence to Galois theory, or even more so to class-field theory.

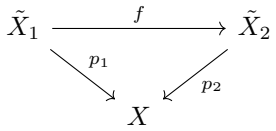
First, we need the following definition

Definition 2.3.5: semi-locally simply connected

A space X is said to be **semi-locally simply connected** if for every point $x \in X$, there exists an open $x \in U$ such that the inclusion induced map $\pi_1(U, x) \to \pi_1(X, x)$ is trivial.

Note that in the above definition $\pi_1(U, x)$ need not itself be trivial. What the definition says is that loops in U at x are contractible in X, but the contraction need not all happen inside of U.

Next, when talking about isomorphisms of covering spaces, we of course mean a map $f_1 : \tilde{X_1} \to \tilde{X_2}$ such that the following commutes:



This allows us to state

Theorem 2.3.6: Classification of covering spaces

Let X be path-connected, locally path-connected and semi-locally simply connected. Sending a path-connected covering space $p : \pi_1(\tilde{X}, \tilde{x_0})$ to the subgroup $p_*(\pi_1(\tilde{X}, \tilde{x_0})) \subset \pi_1(X, x_0)$ induces a bijection between isomorphism-classes of base-point preserving path-connected covering-spaces and subgroups of $\pi_1(X, x_0)$.
Ignoring base-points, the association induces a bijection between path-connected covering spaces, and conjugacy classes of subgroups of $\pi_1(X, x_0)$.

Proof. The main idea is to construct a simply-connected covering space just from X and $\pi_1(X, x_0)$. This space is constructed as

$$\tilde{X} := \{[\gamma] \mid \gamma \text{ a path starting in } x_0\},$$

with a topology defined as follows: First, let $\mathcal{U}$ be the collection of path-connected open sets such that $\pi_1(U) \to \pi_1(X)$ is trivial. The assumptions means that this is a basis of the topology on X. For each $U \in \mathcal{U}$, we can define open sets of the form

$$U_{[\gamma]} := \{[\gamma\eta] \mid \eta \text{ is a path in } U \text{ with } \gamma(1) = \eta(0)\}.$$

This forms a basis of a topology on $\tilde{X}$. From this, most of the stuff follows surprisingly smoothly by using the definitions (see pages 63 - 68 in Hatcher). □

Of course, the above implies the existence of a unique (up to isomorphism) path-connected cover of X corresponding to the trivial subgroup (that would be the simply connected cover that was constructed in the proof). This is called the **universal cover** of X.

Next, we go on to describe deck-transformations. Under composition, these form a group, which is really the analogy of Galois groups.

Definition 2.3.7: Deck transformation

An automorphism of a covering space $\tilde{X} \to X$ is called a **deck transformation**. The group of all deck transformations under composition is denoted $G(\tilde{X}/X)$.

We need another definition whose name will be immidiately justified.

Definition 2.3.8: Normal covering (also called Regularing covering)

A covering space $p : \tilde{X} \to X$ is called **normal** (or sometimes **regular**) if for each $x \in X$ and each pair of lifts $\tilde{x}_1, \tilde{x_2}$ of x, there exists a deck transformation taking $\tilde{x_1}$ to $\tilde{x}_2$.

Proposition 2.3.9

Let X be path-connected and locally path-connected, and let H be the subgroup of $\pi_1(X, x_0)$ corresponding to a path-connected covering space $\tilde{X}$. Then

- The covering is normal if and only if $H < \pi_1(X, x_0)$ is normal.
- We have $G(\tilde{X}/X) \cong N(H)/H$, where $N(H)$ is the normalizer of H.

In particular, the universal cover of X satisfies $G(\tilde{X}/X) = \pi_1(X, x_0)$.

Proof. It is easy to see that changing basepoints of a cover correspond to conjugating H by an element of $\pi_1(X, x_0)$ (one that lifts to a path between the two choices of basepoint of $\tilde{X}$). Thus, an element $[\gamma] \in \pi_1(X, x_0)$ (which lifts to a path from $\tilde{x_0}$ to $\tilde{x_1}$) is in the normalizer of H if and only if $p_*(\pi_1(\tilde{X}, \tilde{x_0})) = p_*(\pi_1(\tilde{X}, \tilde{x_1}))$. Now apply the lifting criterion to see that this condition is equivalent to the existence of a deck transformation from that takes $\tilde{x_0}$ to $\tilde{x_1}$. The first point follows.

The second point is then obvious, by considering the map $N(H) \to G(\tilde{X}/X)$ given by sending a an element of the normalizer of H to the corresponding deck-transformation. Of course, elements of H act trivially, since they lift to loops. □

A lot of this material on covering spaces is analogous to Galois theory. We can think of a cover as an extension of its base space, and of deck transformations as symmetries that fix the base. The following table gives an informal dictionary between covering space theory and Galois theory. Pushing this idea to the fullest leads to the **Étale fundamental group**, which unifies these two viewpoints. Maybe in a later note, we can dive into this topic, but for now the following table will do:

Covering space theory	**Galois theory**
The base space X.	The base field K.
A connected covering space $\tilde{X} \to X$.	A separable algebraic field extension L/K.
The number of sheets of a finite cover: how many points lie over a point of X.	The degree $[L : K]$ of a finite extension: how large the extension is over K.
The universal cover $U \to X$: a simply connected space covering every connected cover of X.	The separable closure K^{sep}/K: a field containing a copy of every separable algebraic extension of K.
The fundamental group $\pi_1(X, x_0) \cong G(U/X)$ controls the covering spaces of X.	The absolute Galois group $\text{Gal}(K^{\text{sep}}/K)$ controls the separable extensions of K.
The deck group $G(\tilde{X}/X)$: symmetries of the cover that leave its projection to X unchanged.	The automorphism group $\text{Aut}(L/K)$: symmetries of L that fix every element of K.
A normal (regular) cover: its deck transformations can move any point of a fibre to any other.	A Galois extension: all the conjugates stay inside the extension, giving the full collection of symmetries.
A cover corresponds to the subgroup $H = p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. Its elements are the loops whose lifts close up.	An intermediate field L corresponds to $\text{Gal}(K^{\text{sep}}/L)$, the automorphisms that fix L pointwise.
Smaller subgroups give larger covers, with more sheets. Normal subgroups give normal covers.	Smaller fixing subgroups give larger fields. Normal subgroups give Galois extensions.
An intermediate cover gives a tower $\tilde{X} \to Y \to X$.	An intermediate field gives a tower $K \subseteq F \subseteq L$. Notice the reversal of arrows.
Choosing $\tilde{x}_0$ in the fibre picks out an actual subgroup; changing this choice conjugates the subgroup.	Choosing an embedding $L \hookrightarrow K^{\text{sep}}$ picks out a particular copy of L; changing it conjugates the fixing subgroup.
Lifting loops permutes the sheets of a cover.	Galois automorphisms permute the embeddings of an extension, or the roots of a defining polynomial.

3 Homology

In this chapter, we define and discuss the basic properties of simplicial and singular homology. We can be quite quick in the discussion of abstract machinery, since they are covered in other notes, but arguably, these homology theories are the simplest to understand, and thus their definition is nice for intuition (even if, as Hatcher nicely puts it, the properties are the most important points for working with it in practice).

3.1 Simplicial and Singular Homology

Recall the following

Definition 3.1.1: n-simplex, faces, boundaries

An **n-simplex** is defined as

$$\Delta^n := \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_i t_i = 1, t_i \geq 0\},$$

thought of as an n-dimensional triangle, together with an ordering on the vertices $[v_0, \dots, v_n]$, according to the index (this gives an orientation of each edge $[v_i, v_j]$). A **face** of Δ^n, is the Δ^{n-1} obtained by deleting one of the verteces, written $[v_0, \dots, \hat{v_i}, \dots, v_n]$. The boundary $\delta\Delta^n$ is the union of all the faces, while the **open n-simplex** is $\mathring{\Delta}^n := \Delta^n - \delta\Delta^n$.

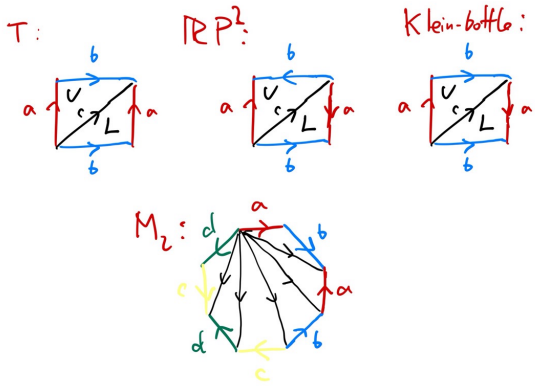
Such simplexes are used to define both simplicial and singular homology. For simplicial homology specifically, we require the following.

Definition 3.1.2: Δ-complex

A **Δ-complex** structure on a topological space, is defined as a collection of maps $\sigma_\alpha : \Delta \to X$ such that the restriction of each map $\sigma_\alpha : \mathring{\Delta}^n \to X$ are injective and non-overlapping (i.e. each $x \in X$ is in the image of exactly one map), the restriction of each $\sigma_\alpha : \Delta^n \to X$ to one of the faces is one of the other maps $\sigma_\beta : \Delta^{n-1} \to X$, and finally, such that $A \subset X$ is open iff. $\sigma_\alpha^{-1}(A)$ is open.

Every Δ-complex is clearly Haussdorff by the last condition. One can even show (I suppose this is quite intuitive...) that they are CW-complexes, so they are not very generic, but they provide good intuition anyway for what is coming.

The following figure shows three simple examples of Δ-complexes, that provide for important calculations later



The bottom example is the orientable surface of genus 2 ("2-holed donut"). It generalizes to genus g, by creating a regular $4g$-gon, and identifying the edges according to the word

$$a_1 b_1 a_1^{-1} b_1^{-1} \dots a_g b_g a_g^{-1} b_g^{-1}.$$

Let $\Delta_n(X)$ be the free abelian group with generators all of the (interiors of the images of) n-simplecies. In Hatcher, these images are written as $e^n_\alpha := \sigma_\alpha(\mathring{\Delta}^n)$, i.e. so that $\Delta_n(X) := \bigoplus_\alpha \mathbb{Z}e^n_\alpha$. The boundary map is defined (on simplecies) by sending it to the alternating sum of its faces (the signs are philosophically supposed to take the orientation into account)

$$\delta_n : \Delta_n(X) \to \Delta_{n-1}(X)$$
$$\delta_n(\sigma_\alpha) = \sum_{i=0}^{n} (-1)^i \sigma_\alpha \mid_{[v_0,\dots,\widehat{v_i},\dots,v_n]} .$$

By direct computation, it one can see that $\delta_{n-1}\delta_n$ is always zero, and thus defines

Definition 3.1.3: Simplicial homology

Simplicial homology is defined as the homology groups of the chain complex $\Delta_\bullet(X)$, i.e.

$$H^\Delta_n(X) = \ker \delta_n / \operatorname{im} \delta_{n+1}.$$

We can now compute some singular homology groups directly from the definition.

Example 3.1.4

For all three examples in the figure from before, $T, \mathbb{RP}^2, K$ (K for Klein-bottle), we have only three non-trivial Δ^n:

$$\cdots \to 0 \to \Delta_2 \to \Delta_1 \to \Delta_0 \to 0$$

In all three cases, the 1-simplexes are labeled a, b, c, and 2-simplexes U, L. The number of vertecies is not the same: we have the same situation for T and K (though the maps themselves of course differ):

$$\cdots \to 0 \to \mathbb{Z}^2 \to \mathbb{Z}^3 \to \mathbb{Z} \to 0$$

while $\mathbb{RP}^2$ is built from two 0-simplexes (vertecies), so the situation is

$$\cdots \to 0 \to \mathbb{Z}^2 \to \mathbb{Z}^3 \to \mathbb{Z}^2 \to 0$$

From explicitly computing the boundary-maps, we can also compute homology groups. We start with T: Clearly, $\delta_1 = 0$, so we only really need to figure out δ_2. Choosing an orientation on U and V (for instance according to the direction of c, we get $\delta_2 U = c - b - a = \delta_2 L$. Choosing the basis for $\Delta^1(X)$ as $[a, b, c - a - b]$, the homology groups become obvious:

$$H^\Delta_k(T) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Next, for K, we again only need to figure out δ_2. We see that in this case, $\delta_2 U = c - b - a$, while $\delta_2 L = c + a - b$. These are clearly linearly independent, and non-zero, so H^2 must vanish. Choosing the basis $[a, b, c]$, we have $H^1(K) = \mathbb{Z}^3/\Lambda$, where Λ is the lattice spanned with basis

$$\begin{pmatrix} -1 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix}$$

from which the SNF gives the homology groups:

$$H_k^\Delta(K) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Finally, for $\mathbb{RP}^2$, lets call the vertices v, w. We have $\delta_1 a = \delta_1 b = \delta_1 c = v - w$, which gives H^0. Further, $\delta_1 U = c + b - c$, while $\delta_1 L = c + a - b$ which is again injective. Putting the maps together, it is not hard to see

$$H_k^\Delta(\mathbb{RP}^2) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

The definition of singular homology is almost as easy, although harder to visualize due to the group of n-chains being so hilariously big.

Definition 3.1.5: singular homology

A **singular n-simplex** is simply a continuous map $\sigma : \Delta^n \to X$. Let $C_n(X)$ be the free abelian group generated by all the singular n-chains. An element of $C_n(X)$ is called a **singular n-chain**. The boundary maps $\partial_n : C_n(X) \to C_{n-1}(X)$ defined analogusly as before, i.e.

$$\partial_n(\sigma) = \sum_{i=0}^{n} (-1)^i \sigma \mid_{[v_0,\ldots,\widehat{v_i},\ldots,v_n]}$$

turn $C_\bullet(X)$ into a chain complex, and we can again define the **singular homology groups** as

$$H_n(X) := \ker \partial_n / \operatorname{im} \partial_{n+1}.$$

Directly from looking at the group of n-chains, one can make three elementay observations:

- Let X decompose into path-components X_α. Then $H_n(X) \cong \oplus_\alpha H_n(X_\alpha)$, since the image of a singular n-simplex is obviously path-connected.
- If X is non-empty and path-connected, then $H_0(X) = \mathbb{Z}$, which follows from the isomorphism $H_0(X) \to \mathbb{Z}; \sum_i n_i x_0 \to \sum_i n_i$.
- If X is (homotopy equivalent to) a point, then $H(X) = 0$ for $n > 0$. Ignoring the parenthesis, which follows from Theorem 3.1.7, the chain complex $C_\bullet(X)$ for X a point is particularily simple: there is a unique n-simplex for every n, and the boundary maps send the n-simplex to sums of $n+1$ $n-1$-simplexes with alternating sign, and are therefore trivial for odd n, and isomorphisms for even n.

Allowing "-1-simplecies" (I guess thought of as a function $\varnothing \to X$), one give the following

Definition 3.1.6: Reduced homology

The **reduced homology group** $\tilde{H}_n(X)$ are the homology groups of the augmented chain complex

$$\cdots \to C_1(X) \to C_0(X) \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(\sum_i n_i x_i) = \sum_i n_i$.

Given a continuous function $f : X \to Y$, the corresponding chain map is denoted

$$f_\sharp : C_\bullet(X) \to C_\bullet(Y)$$
$$f_\sharp(\sigma) = f \circ \sigma$$

and the corresponding map in homology is denoted $f_* : H_n(X) \to H_n(Y)$.

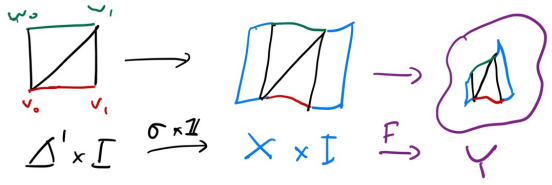
Theorem 3.1.7

If two maps $f \simeq g : X \to Y$ are homotopic, then their induced maps in homology are the same $f_* = g_* : H_n(X) \to H_n(Y)$. Consequently, if $f : X \to Y$ is a homotopy equivalence, then f_* is an isomorphism for all n.

Proof. Of course, the goal is to show that a homotopy induces a chain homotopy between $f_\sharp$ and $g_\sharp$, i.e. maps P, such that $f_\sharp - g_\sharp = \partial P + P\partial$ for all n in the diagram

$$\begin{array}{ccccccccc}
\cdots & \xrightarrow{\partial} & C_{n+1}(X) & \xrightarrow{\partial} & C_n(X) & \xrightarrow{\partial} & C_{n-1}(X) & \xrightarrow{\partial} & \cdots \\
 & \swarrow^{P} & \downarrow{\scriptstyle f_\sharp}\downarrow{\scriptstyle g_\sharp} & \swarrow^{P} & \downarrow{\scriptstyle f_\sharp}\downarrow{\scriptstyle g_\sharp} & \swarrow^{P} & \downarrow{\scriptstyle f_\sharp}\downarrow{\scriptstyle g_\sharp} & \swarrow^{P} & \\
\cdots & \xrightarrow{\partial} & C_{n+1}(Y) & \xrightarrow{\partial} & C_n(Y) & \xrightarrow{\partial} & C_{n-1}(Y) & \xrightarrow{\partial} & \cdots
\end{array}$$

The map P is called the prism operator, and is based on the idea of subdividing $\Delta^n \times I$ into a bunch of Δ^{n+1} simplicies. The following picture shows the idea for $n = 1$ (recall that we have to choose the signs so that the prism operator satisfies the chain homotopi condition for f, g), from which it is pretty easy to derive the general formula.



i.e., given a homotopy $F : X \times I \to Y$ from f to g, one defines

$$P : C_n(X) \to C_{n+1}(Y)$$
$$P(\sigma) = \sum_{i=0}^{n} (-1)^i F \circ (\sigma \times \mathbf{1}) \mid_{[v_0,\ldots,v_i,w_i,\ldots,w_n]}$$

and in the case of $n = 1$, it is easy to see that the relation is satisfied, since all of the black edges cancel. Some tedious algebraic manipulation shows that it holds in general. □

3.2 Excision

We start with

Definition 3.2.1

Let $A \subseteq X$ be a subspace. The group of **relative n-chains** is $C_n(X, A) := C_n(X)/C_n(A)$. Since the boundary map takes $C_n(A)$ to $C_{n-1}(A)$, we may also define the **relative homology groups** $H_n(X, A)$ to be the homology groups of the chain complex $C_\bullet(X, A)$.

Thus, for any subspace $A \subseteq X$, we have an exact sequence of chain complexes

$$0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$$

and subsequently, we have a long-exact sequence in homology

$$\ldots \longrightarrow H_{n+1}(X, A) \xrightarrow{\delta} H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\delta} H_{n-1}(A) \longrightarrow \ldots$$

The snake-lemma maps δ are particularily easily to compute: First, note that the elements of $H_n(X, A)$ can be represented by **relative cycles**, i.e. n-chains $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$. And for relative cycles, the map δ is literally just taking α to $[\partial\alpha] \in H_{n-1}(A)$.

There is also a corresponding long exact sequence for a triple $B \subseteq A \subseteq X$, essentially formed by quotionting by $C_\bullet(B)$ everywhere, i.e.

$$0 \to C_\bullet(A, B) \to C_\bullet(X, B) \to C_\bullet(X, A) \to 0$$

inducing

$$\ldots \xrightarrow{\delta} H_n(A, B) \xrightarrow{i_*} H_n(X, B) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\delta} H_{n-1}(A, B) \longrightarrow \ldots$$

We now get to another big theorem, which is about relative homology groups, known as the excision theorem. It states that when considering relative homology, we may exorcise subsets Z lying inside the subsets A that we are quotienting out by.

Theorem 3.2.2: Excision

Given subsets $Z \subseteq A \subseteq X$ such that $\overline{Z} \subseteq \text{int}(A)$, the inclusion $i : (X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces an isomorphism in homology $i_* : H_k(X \setminus Z, A \setminus Z) \xrightarrow{\sim} H_k(X, A)$.

Proof. The proof of this theorem is rather long. The essential lemma is even more general: Let $\mathcal{U} = U_j$ denote an open cover of X, and let $C_\bullet^{\mathcal{U}}(X) := \sum_j C_\bullet(U_j) \subseteq C_\bullet(X)$ (note: not direct sum, because the U_j generally overlap). Then the inclusion

$$i : C_\bullet^{\mathcal{U}}(X) \hookrightarrow C_\bullet(X)$$

is a chain homotopy equivalence! The proof of this is through the technique called barycentric subdivision: essentially the idea is to subdivide simplicies into smaller and smaller simplicies such that they are eventually all contained in a single component U_j.

Anyway, from this lemma the theorem follows by taking the open cover $\{\text{int}(A), X \setminus \overline{Z}\}$. □

There is another equivalent formulation which is often more useful.

Corollary 3.2.3

Let $A, B \subseteq X$ be subsets such that $\text{int}(A) \cup \text{int}(B) = X$, the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces an isomorphism $H_k(B, A \cap B) \xrightarrow{\sim} H_k(X, A)$ in homology.

Proof. Set $B = X \setminus Z$, and $A = X \setminus B$, and apply excision. □

Definition 3.2.4: Good pair

A **good pair** (X, A) is a space X and a nonempty closed subset $A \subseteq X$ that is a deformation retract of some neighborhood in X

The reason for pointing out such pairs is that excision makes relative homology particularily nice to describe for good pairs.

Proposition 3.2.5

For good pairs (X, A), the quotient map $q : X \to X/A$ induces an isomorphism $q_* : H_k(X, A) \xrightarrow{\sim} H_k(X/A, A/A) \simeq \tilde{H}_k(X/A)$ in homology.

Proof. See the book for a full proof. The idea is to thinken to a neighboorhood $A \subseteq V$ which deformation retracts onto A, which then gives $H_k(X, A) \simeq H_k(X, V)$ by the long-exact sequence in homology for $A \subseteq V \subseteq X$. Excision then gives $H_k(X, V) \simeq H_k(X \setminus A, V \setminus A)$. But now the restricted quotient map $q : X \setminus A \to X/A \setminus A/A$ is a homeomorphism, which finishes the job. □

The tools at this point are enough to compute many homology groups and get quick proofs of cool, non-trivial facts. We give one example here, see the book for more.

Theorem 3.2.6

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be nonempty, open and homeomorphic. Then $n = m$.

Proof. Take $x \in U$. By excision, we have $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \simeq H_k(U, U - x)$. From the long exact sequence, we get that $H_k(\mathbb{R}^n, \mathbb{R}^n \setminus \{x\}) \simeq H_{k-1}(\mathbb{R}^n \setminus \{x\}) \simeq H_{k-1}(S^{n-1})$ (the last isomorphism is from a deformation retract). But now its obvious, by the same reasoning for V. □

We finally state a theorem for our sanity, but without proof:

Theorem 3.2.7

Let X be a Δ-complex, and $A \subseteq X$ a subcomplex. The obvious homomorphisms

$$H_k^\Delta(X, A) \to H_k(X, A).$$

are isomorphisms for all k.

Proof. I don't really care that much. □

Next, the following is an easy corollary of the long-exact sequence in homology, and the proof of excision, but it is computationally very useful. It is also a homological version of the Van-Kampen theorem.

Theorem 3.2.8: Mayer-Vietoris Sequence

Let $A, B \subseteq X$ be such that $\text{int}A \cup \text{int}(B) = X$. Then there is a long exact sequence

$$\ldots \longrightarrow H_n(A \cap B) \longrightarrow H_n(A) \oplus H_n(B) \longrightarrow H_n(X) \longrightarrow H_{n-1}(A \cap B) \longrightarrow \ldots$$

Proof. In the proof of excision, we mentioned that the key idea is that the obvious map

$$(C_\bullet(A + B) :=) C_\bullet(A) + C_\bullet(B) \to C_\bullet(X)$$

is a chain homotopy. Then the obvious exact sequence of chain complexes

$$0 \to C_\bullet(A \cap B) \to C_\bullet(A) \oplus C_\bullet(B) \to C_\bullet(A + B) \to 0$$

gives the theorem. □

3.3 Degree, and cellular homology.

TODO.

3.4 Eilendberg-Steenrod axioms

There are of course many many homology theories. The following axiomatization says what they should at least have in common.

Definition 3.4.1: Eilenberg-Steenrod axioms for homology

A homology theory is a sequence of functors H_n from the category of pairs (X, A) of topological spaces, to the category of abelian groups, together with natural transformations $\delta_n : H_n(X, A) \to H_{n-1}(A, \varnothing)$ such that:

Homotopy: Homotopic maps $f \simeq g : (X, A) \to (Y, B)$ induce the same maps in homology.

Excision: The functors H_n satisfy the excision theorem.

Dimension: $H_n(x) = 0$ for all $n \neq 0$.

Additivity: If $X = \sqcup_\alpha X_\alpha$, then $H_n(X) \simeq \oplus_\alpha H_n(X_\alpha)$.

Exactness: Each pair (X, A) induces a long exact sequence in homology.

Of course, this seems like a rather restrictive definition, because if obviously works just as well when H_n takes values in R-modules rather than simply abelian groups (or indeed, any other abelian category).

So let us indeed define right away

Definition 3.4.2: Singular homology with R-coefficients

For a pair (X, A), and a ring R, we can form the chain complex

$$C_\bullet(X, A; R) := C_\bullet(X, A) \otimes_{\mathbb{Z}} R$$

and define the **homology groups with coefficients in** R $H_n(X, A; R)$ to be the homology groups of $C_\bullet(X, A; R)$.

The above can actually be defined for arbitrary abelian groups, not only rings, but the advantage of doing it with a ring is that then the homology groups are naturally R-modules.

4 Cohomology

In this section, we define cohomology, and show that it is determined fully by homology with $\mathbb{Z}$ coefficients by the universal coefficient theorem. We discuss the UCT in some detail, as it is a fairly simple, but powerfull construction.

Next, we define the cup product, and show the Künneth formula. Algebraically, this can be seen as a generalization of the UCT, but it also has a much simpler formulation for cellular cohomology at least.

Finally, we discuss and prove Poincaré duality.

4.1 The universal coefficient theorem

Hatcher works with a general abelian group G, but we will be happy to simply use a ring R.

Definition 4.1.1: Cohomology groups

For a singular complex $C_\bullet(X)$, define the **singular cochain complex with coefficients in** R as $C^\bullet(X; R) := \mathrm{Hom}(C_\bullet(X), R)$ (where the homset is as abelian groups). Denoting the boundary maps $\delta := \partial^*$, we define the **singular cohomology groups** as

$$H^n(X) := \ker \delta_n / \operatorname{im} \delta_{n-1}.$$

By definition, the maps $\delta_n := \partial_n$ takes an n-cochain φ to the $n+1$-cochain which takes a simplex $\rho : \Delta^{n+1} \to X$ to

$$\delta\varphi(\rho) = \sum_{i=0}^{n+1} (-1)^i \varphi(\rho\,|_{[v_0,\ldots,\widehat{v_i},\ldots,v_{n+1}]}).$$

From now on, we denote $Z_n := \ker \partial_n, Z^n := \ker \delta_n, B_n := \operatorname{im} \partial_{n+1}, B^n := \operatorname{im} \delta_{n-1}$. The UCT is really a theorem of homological algebra, but it allows us to relate singular cohomology to singular homology, and we can state right away

Theorem 4.1.2: Universal Coefficient theorem

Let $C_\bullet$ be a chain complex of free abelian groups. For any abelian group A, and any n, there is a split-exact sequence

$$0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_\bullet), A) \to H^n(\mathrm{Hom}(C_\bullet, A)) \to \mathrm{Hom}(H_n(C), A) \to 0.$$

Proof. First, note that there is an obvious homomorphism

$$h : H^n(\mathrm{Hom}(C_\bullet, A)) \to \mathrm{Hom}(H_n(C_\bullet), A),$$

which takes a cohomology class $[\varphi] \in H^n(\mathrm{Hom}(C_\bullet, A))$ to $\overline{\varphi_0}$, i.e. the restriction + quotient of $\varphi_n : C_n \to A$ to a map $Z_n/B_n \to A$. This is surjective because for each n, one has a split exact sequence $0 \to Z_n \to C_n \to B_{n-1} \to 0$ (because B_k are all free), and thus there exists a projection $p : C_n \to Z_n$, which means that any map $Z_n \to A$ extends to a map $C_n \xrightarrow{p} Z_n \to A$. This proves not only surjectivity, but we have even just constructed a section of h, and thus the sequence

$$0 \to \ker h \to H^n(\mathrm{Hom}(C_\bullet, A)) \to \mathrm{Hom}(H_n(C_\bullet), A) \to 0,$$

so it remains only to prove $\ker h \simeq \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_\bullet), A)$.

Consider first the exact sequence of chain complexes

$$0 \to Z_\bullet \to C_\bullet \to B_\bullet \to 0$$

which, a little more detailed, at n looks like

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Z_n & \longrightarrow & C_n & \xrightarrow{\partial} & B_{n-1} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle 0} & & \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle 0} & & \\ 0 & \longrightarrow & Z_{n-1} & \longrightarrow & C_{n-1} & \xrightarrow{\partial} & B_{n-2} & \longrightarrow & 0 \end{array}$$

Applying $(-)^* := \mathrm{Hom}(-, A)$ to this still gives an exact sequence of chain complexes,

$$0 \to B^*_\bullet \to C^*_\bullet \to Z^*_\bullet \to 0$$

because all the vertical rows split by the earlier argument. This then gives a long-exact sequence in cohomology

$$\cdots \to B_{n-1} \xrightarrow{\partial^*} H^n(C^*_\bullet; A) \to Z_n \xrightarrow{i_n^*} B_n \to \dots$$

which gives short exact sequences

$$0 \to \mathrm{coker}\, i^*_{n-1} \to H^n(C^*_\bullet, A) \to \ker i^*_n \to 0$$

where $\ker i^*_n \simeq \mathrm{Hom}(H_n(C_\bullet), A)$, which shows that $\ker h \simeq \mathrm{coker}\, i^*_{n-1}$.

Luckily, this is easy to compute, as it comes from a free resultion

$$0 \to B_{n-1} \xrightarrow{i_{n-1}} Z_{n-1} \to H_n(C_\bullet) \to 0,$$

and applying $\mathrm{Hom}(-, A)$ now gives by definition (and the length of the resultion) $\mathrm{coker}\, i^*_{n-1} := \mathrm{Ext}^1_{\mathbb{Z}}(H_n(C_\bullet), A)$. □

In this case we started with a chain complex $C_\bullet$, and applied the functor $\mathrm{Hom}(-, A)$. However, the proof works just as well, starting with a cochain complex, and/or applying $- \otimes A$ instead.

Theorem 4.1.3: Universal Coefficient Theorem - Variations

Let $C_\bullet$ be a chain complex of abelian groups, let $C^\bullet$ be a cochain complex of abelian groups, and let A be an abelian group. There are, for every n exact sequences

$$\begin{aligned}
0 \to H_n(C_\bullet) \otimes A &\to H_n(C_\bullet \otimes A) &&\to \mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(C_\bullet), A) \to 0 \\
0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_\bullet), A) &\to H^n(\mathrm{Hom}(C_\bullet, A)) &&\to \mathrm{Hom}(H_n(C_\bullet), A) \to 0 \\
0 \to H^n(C^\bullet) \otimes A &\to H^n(C^\bullet \otimes A) &&\to \mathrm{Tor}_1^{\mathbb{Z}}(H^{n+1}(C^\bullet), A) \to 0 \\
0 \to \mathrm{Ext}^1_{\mathbb{Z}}(H^{n+1}(C^\bullet), A) &\to H_n(\mathrm{Hom}(C^\bullet, A)) &&\to \mathrm{Hom}(H^n(C^\bullet), A) \to 0
\end{aligned}$$

Proof. Its the same proof as before, just flipping the arrows at the right times etc. □

The assumption that we were working with abelian groups here can of course be generalized to modules over a PID with no trouble. For modules over a general commutative ring, there also exists a generalization, however it is much harder, and spectral sequences play a central role appearently.

We also mention in passing that the morphisms in the UCT are natural (clear from proof), though the splitting is not. This makes it easy to prove e.g. that if a map of chain complexes $f : C_\bullet \to D_\bullet$ induces an isomorphism in all homology groups, it also does so in all cohomology groups. Notice that this statement is about the map itself: even without naturality it is clear that spaces with isomorphic homology must of isomorphic cohomology, but naturality gives the commutativity of the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(C_\bullet), A) & \longrightarrow & H^n(C^*_\bullet) & \longrightarrow & \mathrm{Hom}(H_n(C_\bullet), A) & \longrightarrow & 0 \\
 & & \uparrow{\scriptstyle (f_*)^*} & & \uparrow{\scriptstyle f^*} & & \uparrow{\scriptstyle (f_*)^*} & & \\
0 & \longrightarrow & \mathrm{Ext}^1_{\mathbb{Z}}(H_{n-1}(D_\bullet), A) & \longrightarrow & H^n(D^*_\bullet) & \longrightarrow & \mathrm{Hom}(H_n(D_\bullet), A) & \longrightarrow & 0
\end{array}$$

For actual computations, it is not too hard when the homology groups are finitely generated, using the general rules $\mathrm{Ext}^n_R(H \oplus H', A) \simeq \mathrm{Ext}^n_R(H, A) \oplus \mathrm{Ext}^n_R(H', A)$, $\mathrm{Ext}^n_R(H, A) \simeq 0$ if H is projective, and $\mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \simeq A/nA$.

Largely using the UCT, one proves that all the material from the previous section extends to cohomology without much difficulity (keeping track of which directions stuff go is the hardest part...). This will now be assumed from here on.

4.2 Cup product and Künneth formulas.

One of the reasons that cohomology is often "simpler" than homologoy, is the following

Definition 4.2.1: Cup product

Given cochains $\varphi \in C^k(X; R), \psi \in C^\ell(X; R)$ one defines the **cup product** as the cochain $\varphi \smile \psi \in C^{k+\ell}(X; R)$ with values on singular simplexes

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma\,|_{[v_0,\dots,v_k]})\varphi(\sigma\,|_{[v_k,\dots,v_{k+\ell}]}).$$

Proposition 4.2.2

The cup product induces a well-defined, bilinear map in cohomology

$$\smile: H^k(X;R) \times H^\ell(X;R) \to H^{k+\ell}(X;R).$$

(There are also relative versions).

Proof. The fact that it gives a well defined map follows from the Leibnitz rule

$$\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi,$$

which can be verified by direct computation. Bilinearity is obvious from the definition. □

The cup product gives $H^*(X;R)$ into a ring. More accurately:

Theorem 4.2.3

Let R be a commutative right. The cup-product makes $H^*(-;R)$ a contraviariant functor from topological spaces into graded-commutative R-algebras.

Proof. The functoriality part, given a continuous map $f : X \to Y$, the induced map $f^* : H^k(X;R) \to H^k(Y;R)$ satisfies $f^*(\alpha \smile \beta) = f^*(\alpha) \smile f^*(\beta)$. The graded-commutative part, which means that

$$\alpha \smile \beta = (-1)^{k\ell} \beta \smile \alpha$$

takes substantially more work. Hatcher does it by reversing the order of vertecies on a singular simplex σ, calling that simplex $\bar{\sigma}$, and then showing that the map

$$p : C_n(X) \to C_n(X)$$
$$p(\sigma) = \epsilon_n \bar{\sigma}$$

where $\epsilon_n := (-1)^{n(n+1)/2}$, is a map that is chain homotopic to the identity. From this the theorem follows by comparing $p^*(\varphi \smile \psi)$ and $p^*(\varphi) \smile p^*(\psi)$, since the maps p^* disappear when passing to cohomology.

One can also show it follows from the cross product, using the alternative definition

$$\alpha \smile \beta = \Delta^*(\alpha \times \beta),$$

where $\Delta : X \to X \times X; x \to (x,x)$ is the diagonal, however, it is not really much easier to prove for the cross product. □

To state the Künneth formula, we need the following

Definition 4.2.4: Cross product

Let X, Y be topological spaces, and denote the projection maps from $X \times Y$ by p_1 and p_2. The **cross product** is the map

$$\begin{aligned} \times : H^*(X;R) \times H^*(Y;R) &\to H^*(X \times Y;R) \\ \alpha \times \beta &= p_1^*(\alpha) \smile p_2^*(\beta). \end{aligned}$$

It is obviously bilinear, and thus gives a ring-homomophism

$$H^*(X;R) \otimes_R H^*(Y;R) \to H^*(X \times Y;R),$$

where a tensor product of graded-commutative rings has multiplication defined by

$$(a \otimes b)(c \otimes d) = (-1)^{|b||c|}(ac \times bd).$$

Checking that the above definition makes sense is routine.
We can then state

Theorem 4.2.5: Künneth formula

Let X, Y be topological spaces, and assume that $H^k(Y;R)$ is a finitely generated R-module for all k. Then the cross product

$$H^*(X;R) \otimes_R H^*(Y;R) \to H^*(X \times Y;R)$$

is an isomorphism.

Proof. Hatcher proves it only for CW complexes, but it is true in general. I will wait, or perhaps write it out later. □

TODO: Write out computation of ring structure of the cohomology ring of some projective spaces.

4.3 Some algebra and computations.

A bit of a tricky example of a graded commutative ring is

Definition 4.3.1: Exterior algebras

Let R be a commutative ring with identity and let V be an R-module. The **exterior algebra** of V is defined from the tensor algebra

$$T_R(V) := \bigoplus_{k\geq 0} V^{\otimes_R k}$$

as

$$\Lambda_R(V) := T_R(V)/\langle v \otimes v \mid v \in V\rangle.$$

The multiplication in $\Lambda_R(V)$ is the **wedge product** $a \wedge b$. It inherits a grading from $T_R(V)$. When V is free with basis $\alpha_1, \ldots, \alpha_n$, we denote it by $\Lambda_R[\alpha_1, \ldots, \alpha_n]$, and $\Lambda^k_R(V)$ has a basis by $\alpha_{i_1} \wedge \cdots \wedge \alpha_{i_k}$, $i_1 < \ldots i_k$. In fact, one can be a bit more general in this case, and define the grading by the α_i, for any odd choice of degrees of the generators.

As an example, for the n-torus T^n, we have $H^*(T^n; R) = \Lambda_R[\alpha_1, \ldots, \alpha_n]$ (here, α_i essentially corresponds to the dual basis, when we identify $H^k(T^n; R) \simeq \mathrm{Hom}(H_k(T^n), R)$ using UCT.

4.4 Poincaré duality

We have finally reached "rosinen i pølsa", but before we can state the theorem statement, we need to know what objects we are talking about.

Definition 4.4.1: Manifold

An **n-manifold** is a locally Euclidean, Haussdorff space.

To remind you, locally Euclidean means that every $x \in M$ has an open neighborhood $U \simeq \mathbb{R}^n$.

We use the notation $H_*(X \mid x; R) := H_*(X, X - \{x\})$ for the local (co)homology at x.

Poincare duality involves three ingredients we still need to define: The concept of orientations, the highly related fundamental class $[M] \in H_n(M; R)$, and the cap product. The statement will be that for a *closed, R-orientable n-manifold* M, the map

$$\begin{aligned} D : H^k(M; R) &\to H_{n-k}(M; R) \\ \alpha &\to [M] \frown \alpha \end{aligned}$$

is an isomorphism

We start working towards defining orientations:

Proposition 4.4.2

Let M be an n-manifold. For $x \in M$, we have

$$H_n(M \mid x; R) = R,$$

and 0 for the i-th homology, for any $i \neq n$.

Proof. The definition of a manifold + excision gives $H_i(M \mid x; R) = H_i(\mathbb{R} \mid x; R)$. This can be computed from the long exact sequence in homology:

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & H_i(\mathbb{R}^n; R) & \longrightarrow & H_i(\mathbb{R}^n \mid x; R) & \longrightarrow & H_{i-1}(\mathbb{R}^n - \{x\}; R) & \longrightarrow & H_{i-1}(\mathbb{R}^n) & \longrightarrow & \ldots \\ & & \Downarrow & & & & & & \Downarrow & & \\ & & 0 & & & & & & 0 & & \end{array}$$

which gives $H_i(\mathbb{R}^n \mid x; R) \simeq H_{i-1}(\mathbb{R}^n - \{x\}; R)$, which in turn is isomorphic to $H_{i-1}(S^{n-1}; R)$ since $\mathbb{R}^n - \{x\}$ is homotopic to S^{n-1}. □

By the previous lemma, and the notion of generators of $H_k(S^k; R)$ as a single rotation of S^k, we define

Definition 4.4.3: Orientations

For an n-manifold M, a **local orientation** at x is a choice of generator $\mu_x \in H_n(M \mid x; R)$. An **$R$-orientation** of M is a function $x \to \mu_x$ which assigns to each $x \in M$ a $\mu_x \in H_n(M \mid x; R)$ in a compatible way, i.e. so that there exists a chart $\mathbb{R} \simeq U$ near x, containing an open ball B so that for all $y \in B$, μ_y is the image of a single generator $\mu_B \in H_n(M \mid B; R)$.
If an R-orientation of M exists, we say that M is R-orientable.

Notice how similar the local condition is to the notion of compatible stalks in algebraic geometry (well, you need Poincaré duality to be talking about sections even here, but you get the point).

The next theorem summarizes orientability in terms of homology. To prove it, one proves the following lemma:

Lemma 4.4.4

Let M be an n-manifold, and let $A \subset M$ be a compact subset.

- For any section $x \to \mu_x$ of the covering space M_R of M, there is a unique class $\mu_A \in H_n(M \mid A; R)$, whose image in $H_n(M \mid y; R)$ is μ_y for all $y \in A$.
- $H_i(M \mid A; R) = 0$ for $i > n$.

Proof. The proof is rather long, but introduces important tools, such as the covering space M_R of M defined in terms of local orientations. Will write it up later, TODO. □

In fact, the above lemma is best understood in terms of the so-called **orientation sheaf** Γ_R, defined as the sheafification of

$$U \to H_n(M \mid U; R).$$

We will need it also to prove

Theorem 4.4.5

Let M be a closed, connected n-manifold, and consider the map

$$f_x : H_n(M; R) \to H_n(M \mid x; R) \simeq R.$$

If M is R-orientable, the map f_x is an isomorphism for all x. If M is not R-orientable, the map f_x is injective with image $R[2]$ for all x. In either case, $H_i(M; R) = 0$ for $i > n$.

Proof. Closed manifolds are also compact, so we can choose $A = M$. There is an R-module homomorphism $H_n(M;R) \to \Gamma_R(M); \alpha \to (x \to \alpha_x)$, and part the first bullet of the lemma asserts that this is an isomorphism. The results now follows from the structure of M_R (which we did not yet write up). □

Definition 4.4.6: Fundamental class

Let M be an R-orientable closed, connected n-manifold. A class $[M] \in H_n(M;R)$ which maps to a generator of $H_n(M \mid x;R)$ for all $x \in M$ is called a **fundamental class** of M with coefficients in R (also sometimes referred to as a **orientation class**).

The book also gives a few small corollaries of the theorem.
Now we can define another ingredient.

Definition 4.4.7: Cap product

For a topological space X, and $k \geq \ell$, we define the **cap** product as

$$\begin{aligned} C_k(X;R) \times C^\ell(X;R) &\to C_{k-\ell}(X;R) \\ \sigma \frown \varphi &= \varphi(\sigma\mid_{[v_0,\dots v_\ell]})\sigma\mid_{[v_\ell,\dots,v_k]} \end{aligned}$$

Proposition 4.4.8

The cap product induces a well-defined bilinear map in (co)homology

$$\frown : H_k(X;R) \times H^\ell(X;R) \to H_{k-\ell}(X;R)$$

(there are also relative versions).

Proof. The cap product, like the cup product satisfies a version of the leibnitz rule

$$\partial(\sigma \frown \varphi) = (-1)^\ell(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi),$$

which can again be checked by direct calculation. □

The cap product also satisfies a form of naturality, although it is slightly confusing. The correct notion is that given a map $f : X \to Y$, we have a diagram

$$\begin{array}{ccccc} H_k(X) & \times & H^\ell(X) & \xrightarrow{\frown} & H_{k-\ell}(Y) \\ \downarrow{\scriptstyle f_*} & & {\scriptstyle f^*}\uparrow & & \downarrow{\scriptstyle f_*} \\ H_k(Y) & \times & H^\ell(Y) & \xrightarrow{\frown} & H_{k-\ell}(Y) \end{array}$$

satisfying

$$f_*(\alpha) \cup \varphi = f_*(\alpha \frown f^*(\varphi)).$$

We now seem to have all the tools necessary for stating Poincaré duality, but we actually need another technical definition, because the proof will really be through a slightly more general statement than before.

Definition 4.4.9: Cohomology with compact support

Let $C_c^i(X;R)$ be the subgrouop of $C^i(X;R)$ consisting of cochains φ, for which there exists some compact set $K_\varphi \subset X$ such that φ is zero on every chain in $X - K_\varphi$. These form a subcomplex, and we define the **cohomology with compact support** $H_c^*(X;R)$ to be the homology groups of this complex.

Note that the definition above says that φ is zero on every chain in $X - K$, it does NOT say the much stronger statement that it is non-zero only on K, because this would not be a chain complex.

Proposition 4.4.10

We have

$$\varinjlim H^i(X \mid K;R) \simeq H_c^i(X;R),$$

where the direct limit ranges over the compact subsets of X. In particular, if X is compact, then $H^*(X;R) \simeq H_c^*(X;R)$.

Proof. Filtered colimits are exact. □

Finally, we one can prove

Theorem 4.4.11: Poincaré duality

Let M be an R-oriented n-manifold. The duality map

$$D_M : H_c^k(M;R) \to H_{n-k}(M;R)$$
$$D_M(\varphi) = [M] \frown \varphi$$

is an ismorphism.

Proof. This proof is genuinely quite long and hard, see book. To define the cap-product from cohomology with compact support, use a limit argument, and the naturality property mentioned before. □

5 Exercise Solutions

Exercise 5.0.A (3:2:x)**.** For a closed orientable surface M of genus $g \geq 1$, show that for each nonzero $\alpha \in H^1(M;\mathbb{Z})$ there exists $\beta \in H^1(M;\mathbb{Z})$ with $\alpha\beta \neq 0$. Deduce that M is not homotopy equivalent to a wedge sum $X \vee Y$ of CW complexes with nontrivial reduced homology. Do the same for closed nonorientable surfaces using cohomology with $\mathbb{Z}/2\mathbb{Z}$ coefficients.

Solution.

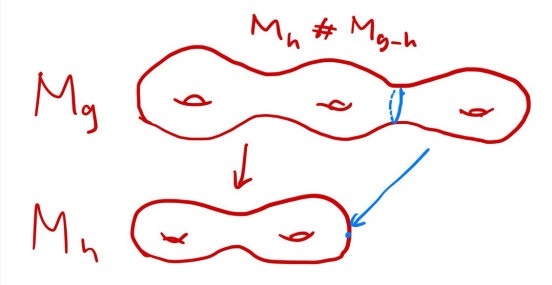
□

Exercise 5.0.B (3.3:11)**.** If M_g denotes the closed orientable surface of genus g show that degree 1 maps $M_g \to M_h$ exists if and only if $g \geq h$.

Solution. $g \geq h$: Write M_g as the connected sum $M_h \# M_{g-h}$. Recall that this means $A := M_h \setminus \mathrm{int}(D^2)$, $B := M_{g-h} \setminus \mathrm{int}(D^2)$ glued along ∂D^2. Then we can construct a map $p : M_g \to M_g/B \cong A/\partial D^2 \cong M_h$. Seing that this is a map of degree one is easy: Recall that we can compute the degree by the diagram

$$\begin{array}{ccc} H_2(M_g;\mathbb{Z}) & \longrightarrow & H_2(M_h;\mathbb{Z}) \\ \downarrow & & \downarrow \\ H_2(M_g \mid p^{-1}(x);\mathbb{Z}) & \longrightarrow & H_2(M_h \mid x;\mathbb{Z}) \end{array}$$

and choosing any x away from the collapsed point shows that this is an obvious isomorphism (use excision on the bottom local homology groups, and p becomes the literal identiy).
To summarize: the map



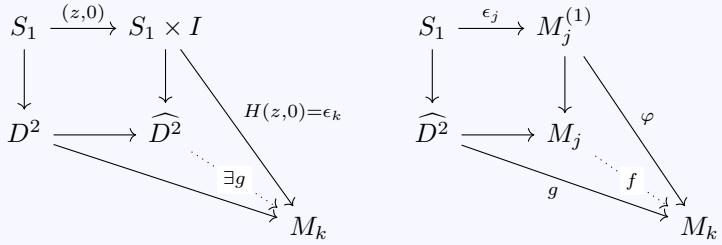
clearly has degree 1.
$g < h$. We could use exercise 3.3:10, but there is a nicer argument. Assume there is a map $f : M_g \to M_h$ of any non-zero degree d. The naturality of the cap product then gives the commutative diagram

$$\begin{array}{ccc} H^1(M_g;\mathbb{Z}) & \xrightarrow{[M_g]\frown -} & H_1(M_g;\mathbb{Z}) \\ {\scriptstyle f^*}\uparrow & & \downarrow{\scriptstyle f_*} \\ H^1(M_h;\mathbb{Z}) & \xrightarrow{d[M_h]\frown -} & H_1(M_h;\mathbb{Z}) \end{array}$$

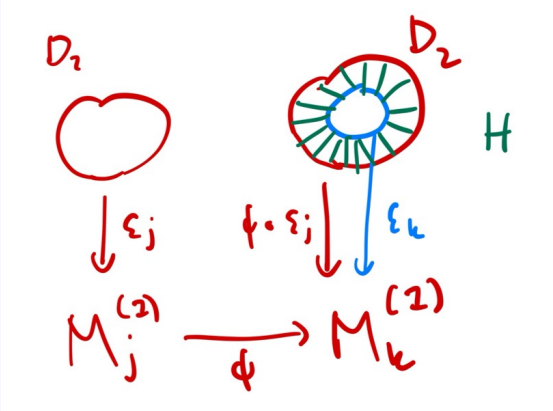
since $f_*([M_g]) = d[M_h]$ by assumption. But this is clearly impossible, because the image of $d[M_h] \frown -$ is d times an isomorphism, hence $d(\mathbb{Z}^{2h})$, and in particular has full rank, so it cannot factor through $\mathbb{Z}^{2g}$. □

Exercise 5.0.C. Show that in a free group F with basis $x_1, \ldots, x_{2k}$, the product of the k commutators $[x_1, x_2] \ldots [x_{2k-1}, 2_{2k}]$ is not equal to the product of $j < k$ commutators $[y_1, y_2] \ldots [y_{2j-1}, y_{2j}]$, for elements $y_i \in F$.

Solution. Of course the idea is to assume that this is possible, and use it to construct a map $M_j \to M_k$. We will use the cellular structure of the two objects.
Recall that for M_g, it has a single 0-cell, $2g$ 1-cells $a_1, \cdots, a_{2g}$, and one 2-cell attached along the word $a_1 a_2 a_1^{-1} a_2^{-1} \cdots a_{2g-1} a_{2g} a_{2g-a}^{-1} a_{2g}^{-1}$. Further, $\pi_1(M_g)$ is the free group with basis the a_i. Lets call the loops in the 1-skeleton $M_j^{(1)}$ as $z_1, \cdots, z_{2j}$, and $M_k^{(1)}$ call $x_1, \cdots, x_{2k}$. Construct a continuous map $\varphi : M_j^{(1)} \to M_k^{(1)}; z_i \to y_i$. Call the attaching maps $\epsilon_j : D^2 \to M_j, \epsilon_k : D^2 \to M_k$. By construction, $\varphi \circ \epsilon_j \simeq \epsilon_k$, because of the assumed equality in $\pi_1(M_k)$. Thus, we have a homotopy $H : S_1 \times I \to M_k$ with $H(z,0) = \epsilon_k$, and $H(z,1) = \varphi \circ \epsilon_j$. Then we have the following two pushout diagrams



Where $\widehat{D^2}$ is read as D^2 with a collar (which is quite obviously the pushout, and also obviously $\widehat{D^2}$ is just "a slightly wider" D^2 whatever that means). The diagram on the left supplies the map g, which, when restricted to $g\,|_{\partial \widehat{D^2}} = H(z,1) = \varphi \circ \epsilon_j$. Thus, the diagram on the right also commutes, and supplies the map f.
This was indeed a bit complicated, but you can visualize the situation by the following figure



Finally, the degree of f is clearly 1, because locally, for instance near the center of D^2, it is a homeomorphism. □